Problem5

October 31, 2026

[ ]:
```python
# Problem 5: Codility "TapeEquilibrium"
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# Date: October 30, 2026


def solution(A):
    # Add up the whole array
    total = sum(A)

    # Running sum of the left part (starts empty)
    left = 0

    # Make a big value so any differences will be smaller
    min_diff = float("inf")

    # Try every split point from 1 to N - 1
    for p in range(1, len(A)):
        left = left + A[p - 1]
        right = total - left
        diff = abs(left - right)

        # Keep smallest difference seen so far
        if diff < min_diff:
            min_diff = diff

    # Return smallest difference found
    return min_diff
```

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References:
[5] Codility Limited, "TapeEquilibrium coding task," Codility. [Online].␣
↪Available: https://app.codility.com/programmers/lessons/3-time_complexity/
↪tape_equilibrium/. [Accessed: Oct. 30, 2026].
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